

Feature Dictionary

FILE 1 — features_master.csv (134 rows × 21 columns)

One row per CFS freight area.

Field	Type	Range / Values	What it means	ML Modeler uses it for
GEO_ID	String	E330000US... (134 unique)	Census identifier for the CFS area. Unique key for joining to shapefiles and all other CSVs.	Join key when merging with distance_matrix.csv
NAME	String	134 unique area names	Human-readable area name. Example: Los Angeles-Long Beach, CA CFS Area; California.	Display label on feature importance charts and classification reports
VAL	Integer	\$12,477M – \$7,963,786M	Total freight value shipped from this area in 2022, in millions of USD. The economic size of the hub. LA-Long Beach = \$7.96 trillion. Highest-weight input feature.	Primary feature for Option C gravity model ($VAL_{origin} \times VAL_{dest}$). Also a feature in Option A classifier.
TON	Integer	4,686 – 3,688,463 (thousand tons)	Total freight weight shipped in 2022, in thousands of short tons. Captures physical volume regardless of dollar value. High TON with low VAL = bulk commodities (grain, sand, fuel).	Feature in Option A classifier. Helps model distinguish high-volume bulk areas from high-value light-freight areas.
val_norm	Float	0.000 – 1.000	Min-max normalized VAL. 1.0 = LA-Long Beach (highest). 0.0 = smallest area. Part of the vulnerability score formula.	DO NOT use as a model feature. It is derived from VAL and makes the model circular.
ton_norm	Float	0.000 – 1.000	Min-max normalized TON. 1.0 = Houston (highest tonnage). Part of the vulnerability score formula.	DO NOT use as a model feature. It is derived from TON and makes the model circular.
vulnerability_score	Float	0.000 – 0.882	Composite risk index = $(val_norm \times 0.60) + (ton_norm \times 0.40)$. Higher = more national disruption if this area fails. Pre-computed target for reference only.	DO NOT use as input feature. Use only to create the risk_tier target label (High/Medium/Low using 90th/75th percentile breaks).
rank	Integer	1 – 134	Ordinal rank by vulnerability score. Rank 1 = most vulnerable (LA-Long Beach). Rank 134 = least vulnerable (Washington DC).	Reference only. Use for sorting output tables and charts. Not a model feature.
is_metro	Integer	0 or 1	1 = metropolitan CFS area (named city hub). 0 = Remainder-of-State zone (non-metropolitan counties grouped together). Derived from NAME field: areas starting with 'Remainder' = 0.	Binary feature in Option A classifier. EDA Finding 5 showed metro and remainder areas have different value/tonnage profiles and should be treated separately.
INTPTLAT	Float	20.80 – 64.20 (decimal degrees)	Latitude of the CFS area's interior point (geographic center). Used for map placement and distance calculations. Hawaii ≈ 20.8°, Alaska ≈ 64.2°.	Used to generate map visualizations of model outputs (risk tier choropleth).
INTPTLON	Float	-157.86 – -69.45 (decimal degrees)	Longitude of the CFS area's interior point. Negative values = west of Greenwich (all U.S. areas). Hawaii ≈ -157.9°, Maine ≈ -69.4°.	Paired with INTPTLAT for map visualization.

nearest_port	String	Port name (26 options)	Name of the closest NTAD intermodal facility to this CFS area, whether seaport or rail hub. Example: Houston = Port of Houston (5.7 miles). Chicago = Chicago Intermodal Hub (0 miles).	Reference for reporting. The distance version (nearest_port_miles) is used in modeling.
nearest_port_type	String	'seaport' or 'rail_intermodal'	Whether the nearest facility is a seaport or a rail intermodal hub. Chicago and Memphis nearest facilities are rail hubs, not seaports.	Contextual reference. Helps explain why inland areas score differently from coastal areas.
nearest_port_miles	Float	0.0 – 2,402.4 miles	Distance in miles from the CFS area's centroid to the nearest NTAD facility (seaport or rail hub). Honolulu = 2,402 miles. Houston = 0 miles (port is adjacent).	Available as a feature but nearest_seaport_miles is preferred for modeling since it specifically measures maritime access.
nearest_seaport	String	Port name (20 seaports)	Name of the nearest seaport specifically (excludes rail hubs). Example: Chicago's nearest seaport is Baltimore (607 miles). Minneapolis's is Baltimore (939 miles).	Reference for output labeling in the risk tier report.
nearest_seaport_miles	Float	0.57 – 2,402.4 miles	Distance to nearest seaport only. This is the critical isolation metric. Chicago = 607 miles (Baltimore). Minneapolis = 939 miles (Baltimore). Denver = 839 miles (Long Beach). High values = inland areas with no maritime alternative.	KEY FEATURE for Option A classifier. High seaport distance + high freight value = highest risk tier. The model learns this combination predicts High-tier areas.
min_distance_to_neighbor_miles	Float	0.0 – 1,634.1 miles	Distance in miles to the nearest OTHER CFS area. Measures how isolated an area is within the freight network. 0 = adjacent CFS areas (Chicago and Remainder of Illinois share a border). Alaska = 1,634 miles to Seattle.	KEY FEATURE for Option A classifier. An area that is far from its nearest peer has fewer rerouting options if it fails. The model uses this as an isolation signal.
num_commodities	Integer	71 – 368 unique codes	Number of distinct commodity codes (COMM) with non-zero freight value shipped from this area. LA-Long Beach ships 368 different commodity types. More diverse areas may be more resilient to single-sector disruptions.	Feature in Option A classifier. Areas with fewer commodity types may be more fragile to sector-specific disruptions (e.g., an area shipping only energy products).
max_comm_val	Integer	\$3,872M – \$1,227,627M	The freight value (\$M) of the single largest commodity shipped from this area. Measures within-area concentration. If one commodity dominates, a sector disruption is as damaging as a geographic one.	Feature in Option A classifier. High max_comm_val relative to total VAL = concentrated area. The ratio max_comm_val/VAL indicates single-commodity dependency.
dominant_commodity	String	Text label (e.g., 'All Commodities', 'Pharmaceutical products')	The COMM_LABEL of the highest-value commodity shipped from this area. Most large areas show 'All Commodities' (the COMM=0 aggregate row). Specific commodities appear for specialized hubs.	Encoded as a numeric category (LabelEncoder) and used as a feature in Option A. Helps the model distinguish pharma hubs from energy hubs from auto hubs.
val_ton_ratio	Float	0.094 – 4.612	Freight value per thousand tons (VAL/TON). High ratio (>2) = high-value light freight such as electronics, pharmaceuticals. Low ratio (<1) = bulk commodities such as grain, coal, gravel. From the data dictionary's recommended derived feature.	Feature in Option A classifier. Helps model distinguish high-value/low-weight areas (electronics risk) from high-weight/low-value areas (infrastructure risk).

FILE 2 — distance_matrix.csv (17,822 rows × 5 columns) ALL CORRIDOR PAIRS

One row per ordered pair of CFS areas. Covers all $134 \times 133 = 17,822$ origin-destination combinations. Every corridor in the U.S. domestic freight network is represented. Distances computed using the haversine formula from CFS area interior point coordinates (INTPTLAT/INTPTLON).

Field	Type	Range / Values	What it means	ML Modeler uses it for
GEO_ID_origin	String	134 unique CFS GEO_IDs	Census identifier of the freight origin area. This is the area whose failure is being modeled when simulating a disruption.	Join key to features_master.csv to retrieve VAL_origin for the gravity model. Group by this field to compute min_distance_to_neighbor_miles.
NAME_origin	String	CFS area name	Human-readable name of the origin area. Included for readability when inspecting the table.	Display labels in gravity model output tables and corridor ranking charts.
GEO_ID_dest	String	133 unique CFS GEO_IDs per origin	Census identifier of the freight destination area. Every area appears as a destination for every other origin.	Join key to features_master.csv to retrieve VAL_dest for the gravity model. Each origin-destination pair becomes one row in the corridor scoring table.
NAME_dest	String	CFS area name	Human-readable name of the destination area.	Display label in top_corridors_gravity.csv output.
distance_miles	Float	0.0 – 5,110.5 miles. Mean = 1,138 miles	Haversine great-circle distance in miles between the centroids of origin and destination CFS areas. LA to Honolulu = 5,110 miles (longest). Adjacent areas = 0 miles. Mean corridor = 1,138 miles.	CORE INPUT for Option C gravity model: $\text{gravity_score} = (\text{VAL_origin} \times \text{VAL_dest}) / \text{distance}^2$. Short + high-value corridor = highest score. Also grouped by GEO_ID_origin (minimum) to produce min_distance_to_neighbor_miles feature for Option A.

FILE 3 — port_proximity.csv (134 rows × 7 columns) — PORT ISOLATION METRICS

One row per CFS area. Answers: how far is each freight zone from the nearest port? Built from 26 NTAD intermodal facilities (20 seaports + 6 rail hubs). The nearest_seaport_miles column is the most analytically important field in this file.

Field	Type	Range / Values	What it means	ML Modeler uses it for
GEO_ID	String	134 unique CFS GEO_IDs	Join key. Matches GEO_ID in features_master.csv and vulnerability_scores.csv.	Join key when merging into the modeling feature table.
NAME	String	CFS area name	Human-readable area name for display.	Display label in model output reports.
nearest_port	String	26 possible facility names	Name of the closest NTAD facility whether seaport or rail hub. Chicago's nearest is Chicago Intermodal Hub (rail). Houston's nearest is Port of Houston (seaport, 5.7 miles).	Reference for output labeling. nearest_port_miles is the numeric version used in modeling.
nearest_port_type	String	'seaport' or 'rail_intermodal'	Facility type. Distinguishes areas near seaports (coastal) from areas near only rail hubs (inland). Chicago and Memphis nearest facility = rail, not seaport.	Contextual. Helps explain model behavior: inland areas near only rail hubs score differently from coastal areas.
nearest_port_miles	Float	0.0 – 2,402.4 miles	Miles to the nearest NTAD facility of any type. Houston = 0.0 (port is within the area). Honolulu = 2,402 miles to Oakland (across the Pacific).	Available as a feature but nearest_seaport_miles is preferred for its specificity.
nearest_seaport	String	20 possible seaport names	Name of the closest seaport specifically. Chicago's nearest seaport is Baltimore (607 miles). Minneapolis = Baltimore (939 miles). Denver = Port of Long Beach (839 miles). This is the most important text field in the file.	Output label: 'If disrupted, the nearest seaport alternative is [nearest_seaport].'
nearest_seaport_miles	Float	0.57 – 2,402.4 miles. Mainland max ≈ 939 miles (Minneapolis)	Distance in miles to the nearest seaport only (excludes rail hubs). The CRITICAL isolation metric. Low value = coastal area with maritime fallback. High value = inland area with no nearby seaport. Chicago = 607 mi. Minneapolis = 939 mi. Denver = 839 mi.	KEY MODEL FEATURE included in features_master.csv. High seaport distance combined with high VAL = High risk tier. The classifier learns this is the strongest predictor of inland vulnerability.

FILE 4 — ntad_ports.csv (26 rows × 5 columns) REFERENCE FACILITY LIST

The 26 NTAD intermodal facilities are used to compute port_proximity.csv. 20 seaports + 6 major intermodal rail hubs. ML MODELER does not use this file directly. TM4 uses it to label network nodes and cite infrastructure in the policymaker brief.

Field	Type	Range / Values	What it means	ML modeler uses it for
port_name	String	26 unique facility names	Official name of the intermodal facility. Examples: Port of Los Angeles, Chicago Intermodal Hub, Memphis Intermodal Hub.	Not used in modeling.
lat	Float	25.774 – 47.599 (decimal degrees)	Latitude of the port or intermodal hub. Used in haversine calculations to produce port_proximity.csv. All ports are on the continental U.S. coast or major inland rail junctions.	Not used directly by ML MODELER.
lon	Float	-122.762 – -71.041 (decimal degrees)	Longitude of the port. Negative = west of Greenwich (all U.S. facilities). West Coast ports cluster around -118 to -122. East Coast ports cluster around -71 to -77.	Not used directly by ML MODELER.
state	String	2-letter state code. 14 states represented	State where the facility is located. California has the most (3: LA, Long Beach, Oakland). Texas has 3 (Houston, Beaumont, Corpus Christi).	Not used in modeling.
type	String	'seaport' or 'rail_intermodal'	Facility classification. Seaports handle oceangoing container ships. Rail intermodal hubs handle domestic container transfers between truck and rail. Chicago handles ~25% of all U.S. rail freight.	Not used in modeling.

FILE 5 — centroids.csv (134 rows × 4 columns) GEOGRAPHIC COORDINATES

One row per CFS area. Contains the interior point latitude and longitude for each of the 134 CFS freight zones. These coordinates were the source for building the distance matrix and port proximity tables. Both ML MODELER and TM4 use this primarily for visualization.

Field	Type	Range / Values	What it means	ML modeler uses it for
GEO_ID	String	134 unique CFS GEO_IDs	Census geographic identifier. Primary join key to all other files in the project.	Join key for map visualization of model outputs.
NAME	String	134 unique area names	Human-readable CFS area name. Example: Houston-The Woodlands, TX CFS Area; Texas.	Map label for the risk tier choropleth visualization.
INTPTLAT	Float	20.797 – 64.201 (decimal degrees). Hawaii = 20.8°. Alaska = 64.2°	Interior point latitude of the CFS area. This is a representative point guaranteed to fall inside the polygon boundary. Metro areas use the city center. Remainder areas use the state geographic center.	Y-axis coordinate for plotting CFS area locations on the risk tier output map. Pair with INTPTLON for geographic scatter plots.
INTPTLON	Float	-157.858 – -69.445 (decimal degrees). Hawaii = -157.9°. Maine = -69.4°	Interior point longitude of the CFS area. All U.S. values are negative (west of Greenwich). West Coast: -117 to -123. East Coast: -69 to -77. Interior: -85 to -105.	X-axis coordinate for geographic plotting. Pair with INTPTLAT.

Critical Rules for ML MODELER

Rule	Why
Never use val_norm, ton_norm, or vulnerability_score as model features	These are computed from VAL and TON. Using them as inputs makes the model circular — it would just re-learn the formula rather than discovering a genuine pattern.
Use vulnerability_scores_corrected.csv not the original	TM2's EDA found the original file inflated values 3.4x due to SCTG double-counting. The ranking is preserved but absolute values must be corrected. File is in 03_outputs/ per the data dictionary folder structure.
Build Option C before Option A	Option C (gravity model) only needs features_master.csv and distance_matrix.csv — both are ready now. Option A (classifier) also needs these files, so building C first confirms the data pipeline works before adding complexity.
Use class_weight='balanced' in the classifier	There are only 14 High-tier areas vs 101 Low-tier areas. Without balancing, the model will predict Low for everything and claim 75% accuracy while being useless for the actual goal.
Use leave-one-out cross-validation not 80/20 split	With only 134 rows, an 80/20 split leaves only 26-27 test rows — too few for a reliable accuracy estimate. Leave-one-out tests every single area and gives a more honest accuracy number.